

SI for *Museum specimens* reveal butterfly size
increases through time but divergent responses to
temperature

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1 SI Tables

Table S1: Details of butterfly species used in this study. Habitat and foodplant specialism status was split into two categories divided specialists (S) and generalists (G). Numbers for each datasets are the post-filtering counts. See code repository for corresponding .csv file.

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Aglaia urticae</i>	Small tortoiseshell	Nymphalidae	G	S	971	34	1: Jun-Aug	May	Not Listed
<i>Aricia agestis</i>	Brown argus	Lycaenidae	S	S	975	42	2:Aug-Oct	July	Not Listed
<i>Celastrina argiolus</i>	Holly blue	Lycaenidae	G	G	303	25	2:Aug-Oct	June	Not Listed
<i>Lasiommata megera</i>	Wall	Nymphalidae	G	G	619	62	2: Aug-Oct	June	Endangered
<i>Leptidea sinapis</i>	Wood White	Pieridae	S	S	791	70	1: Mar-Jun	JulyBefore	Endangered
<i>Lycaena phlaeas</i>	Small copper	Lycaenidae	G	S	2877	65	2:Jul-Oct	June	Not Listed
<i>Pararge aegeria</i>	Speckled wood	Nymphalidae	G	G	471	38	2:Jul-Aug	June	Not Listed
<i>Pieris brassicae</i>	Large white	Pieridae	G	S	344	44	2:Jul-Sep	June	Not Listed
<i>Pieris napi</i>	Green-veined white	Pieridae	G	G	766	NA	2:Jul-Sep	June	Not Listed
<i>Pieris rapae</i>	Cabbage white	Pieridae	G	G	445	NA	2: Jul-Sep	June	Not Listed
<i>Polygonia c-album</i>	Comma	Nymphalidae	G	G	466	39	1: Jun-Aug	June	Not Listed
<i>Polyommatus bellargus</i>	Adonis Blue	Lycaenidae	S	S	4830	63	2: Aug-Sep	July	Vulnerable
<i>Polyommatus icarus</i>	Common blue	Lycaenidae	G	G	3752	63	2:Aug-Sep	July	Not Listed
<i>Aglaia io</i>	Peacock	Nymphalidae	G	S	505	80	Univoltine	June	Not Listed
<i>Anthocharis car-damines</i>	Orange tip	Pieridae	G	G	1661	64	Univoltine	JulyBefore	Not Listed
<i>Apatura iris</i>	Purple emperor	Nymphalidae	S	S	124	NA	Univoltine	March	Not Listed
<i>Aphantopus hyperan-tus</i>	Ringlet	Nymphalidae	G	G	2206	NA	Univoltine	June	Not Listed
<i>Argynnis paphia</i>	Silver washed fritillary	Nymphalidae	G	S	1447	83	Univoltine	June	Not Listed

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Table S1 – *Continued from previous page*

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Boloria euphrosyne</i>	Pearl-bordered Fritillary	Nymphalidae	G	S	1865	NA	Univoltine	April	Vulnerable
<i>Boloria selene</i>	Small Pearl-bordered Fritillary	Nymphalidae	S	S	1925	NA	Univoltine	April	Vulnerable
<i>Callophrys rubi</i>	Green hairstreak	Lycaenidae	G	G	990	NA	Univoltine	JuneBefore	Not Listed
<i>Carterocephalus palaemon</i>	Chequered Skipper	Hesperiidae	S	S	604	NA	Univoltine	AugustBefore	Not Listed
<i>Coenonympha pamphilus</i>	Small Heath	Nymphalidae	G	S	3058	NA	Univoltine	April	Vulnerable
<i>Coenonympha tullia</i>	Large Heath	Nymphalidae	S	S	1409	44	Univoltine	April	Endangered
<i>Cupido minimus</i>	Small Blue	Lycaenidae	G	S	1453	89	Univoltine	JulyBefore	Near Threatened
<i>Erynnis tages</i>	Dingy skipper	Hesperiidae	G	S	856	85	Univoltine	AugustBefore	Not Listed
<i>Euphydryas aurinia</i>	Marsh Fritillary	Nymphalidae	G	S	1926	NA	Univoltine	May	Vulnerable
<i>Fabriciana adippe</i>	High Brown Fritillary	Nymphalidae	G	S	994	NA	Univoltine	May	Endangered
<i>Favonius quercus</i>	Purple hairstreak	Lycaenidae	G	S	494	83	Univoltine	May	Not Listed
<i>Gonepteryx rhamni</i>	Brimstone	Pieridae	G	S	483	81	Univoltine	June	Not Listed
<i>Hamearis lucina</i>	Duke of Burgundy	Riodinidae	S	S	951	NA	Univoltine	JulyBefore	Vulnerable
<i>Hesperia comma</i>	Silver-spotted Skipper	Hesperiidae	S	S	646	103	Univoltine	June	Vulnerable
<i>Hipparchia semele</i>	Grayling	Nymphalidae	S	G	1654	NA	Univoltine	April	Endangered
<i>Limenitis camilla</i>	White Admiral	Nymphalidae	S	S	761	97	Univoltine	May	Vulnerable
<i>Maniola jurtina</i>	Meadow brown	Nymphalidae	G	G	4730	70	Univoltine	April	Not Listed
<i>Melanargia galathea</i>	Marbled white	Nymphalidae	G	G	2169	101	Univoltine	May	Not Listed
<i>Melitaea athalia</i>	Heath Fritillary	Nymphalidae	G	G	1619	NA	Univoltine	March	Endangered
<i>Melitaea cinxia</i>	Glanville Fritillary	Nymphalidae	S	S	579	NA	Univoltine	March	Endangered
<i>Nymphalis polychloros</i>	Large Tortoiseshell	Nymphalidae	G	G	197	NA	Univoltine	June	Regionally Extinct

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Table S1 – *Continued from previous page*

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Ochlodes sylvanus</i>	Large skipper	Hesperiidae	G	G	1029	NA	Univoltine	May	Not Listed
<i>Papilio machaon</i>	Swallowtail	Papilionidae	S	S	166	54	Univoltine	JulyBefore	Vulnerable
<i>Plebejus argus</i>	Silver-studded Blue	Lycaenidae	S	G	7105	NA	Univoltine	May	Vulnerable
<i>Polyommatus coridon</i>	Chalk Hill Blue	Lycaenidae	G	S	19584	NA	Univoltine	June	Vulnerable
<i>Pyrgus malvae</i>	Grizzled Skipper	Hesperiidae	G	G	1285	108	Univoltine	July	Vulnerable
<i>Pyronia tithonus</i>	Gatekeeper	Nymphalidae	G	S	2313	125	Univoltine	June	Not Listed
<i>Satyrrium pruni</i>	Black Hairstreak	Lycaenidae	S	S	377	NA	Univoltine	May	Endangered
<i>Satyrrium w-album</i>	White-letter Hairstreak	Lycaenidae	S	S	346	NA	Univoltine	May	Vulnerable
<i>Speyeria aglaja</i>	Dark Green Fritillary	Nymphalidae	G	S	1028	51	Univoltine	May	Near Threat- ened
<i>Thecla betulae</i>	Brown Hairstreak	Lycaenidae	G	S	165	NA	Univoltine	June	Vulnerable
<i>Thymelicus acteon</i>	Lulworth Skipper	Hesperiidae	S	S	544	NA	Univoltine	May	Near Threat- ened
<i>Thymelicus lineola</i>	Essex skipper	Hesperiidae	G	G	707	NA	Univoltine	May	Not Listed
<i>Thymelicus sylvestris</i>	Small skipper	Hesperiidae	G	G	918	98	Univoltine	May	Not Listed

UK Red List Status	Size vs Late-Larval Temperature Response		
	–	0	+
Unlisted	10	12	4
Red Listed	5	9	12

Table S2: Categorisation of Temperatures – Size trends by British Red list status (Unlisted versus either *Regionally Extinct*, *Endangered*, *Vulnerable*, or *Near-Threatened*) Temperature responses were categorised into positive, negative or not-detectable based on whether the 90% posterior support of that species’ fitted coefficient straddles zero. (χ^2 -test with Yates’ continuity correction $\chi^2 = 6.0952$, $df = 2$, p-value = 0.047)

2 SI Figures

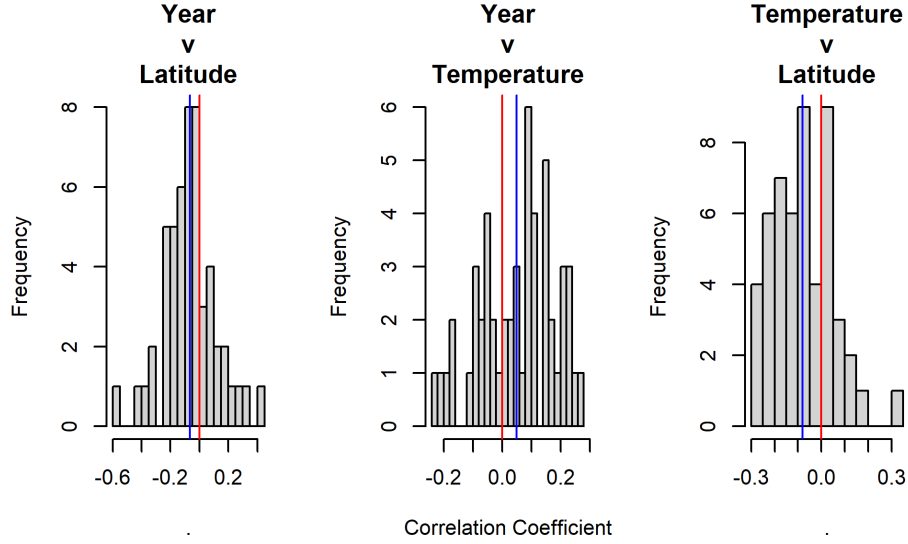


Figure S1: Distribution of within species-level correlations between key predictors in the records in the NHM dataset. Red line shows 0, blue line shows median. There is no consistent geographic trend through time. Temperature tended to be positively correlated with year, but the trend is very noisy. Likewise, there is an overall negative correlation between temperature and latitude, but is neither strong or consistent.

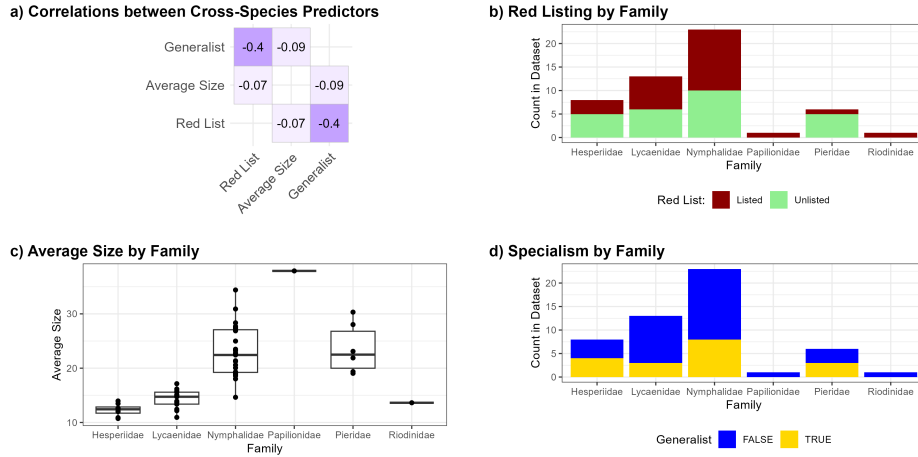


Figure S2: Associations between cross-species predictors. (A) Correlations among the three principal predictors. (B–D) Relationships between taxonomic family and other species-level predictors. Families vary substantially in average size, but are otherwise approximately evenly distributed across predictor space.

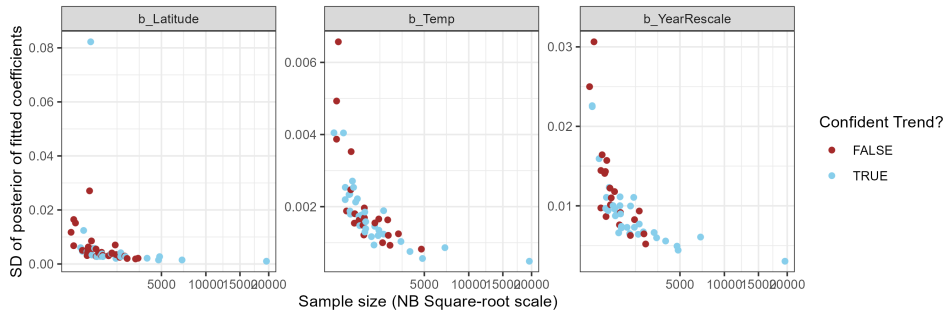


Figure S3: Relationship between sample size, confidence of posterior estimate and identification of species-level trends in the NHM dataset. Precision of posterior estimates was strongly related to the sample size for each species, but the identification of confident trends was less dependent on sample size.

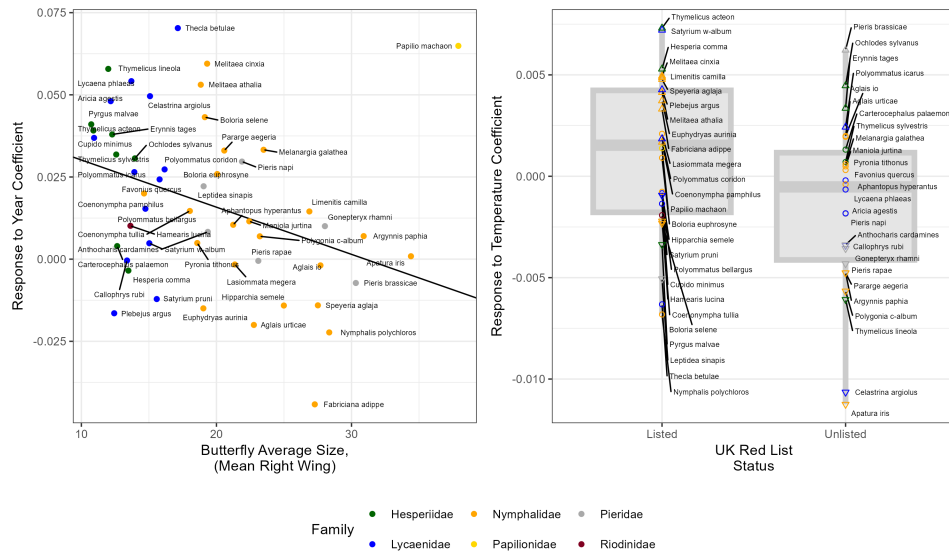


Figure S4: Main text Figure 4 with species labels.

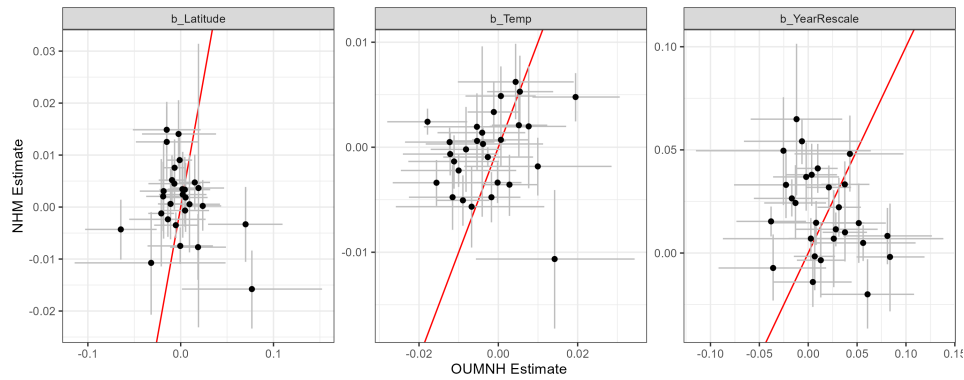
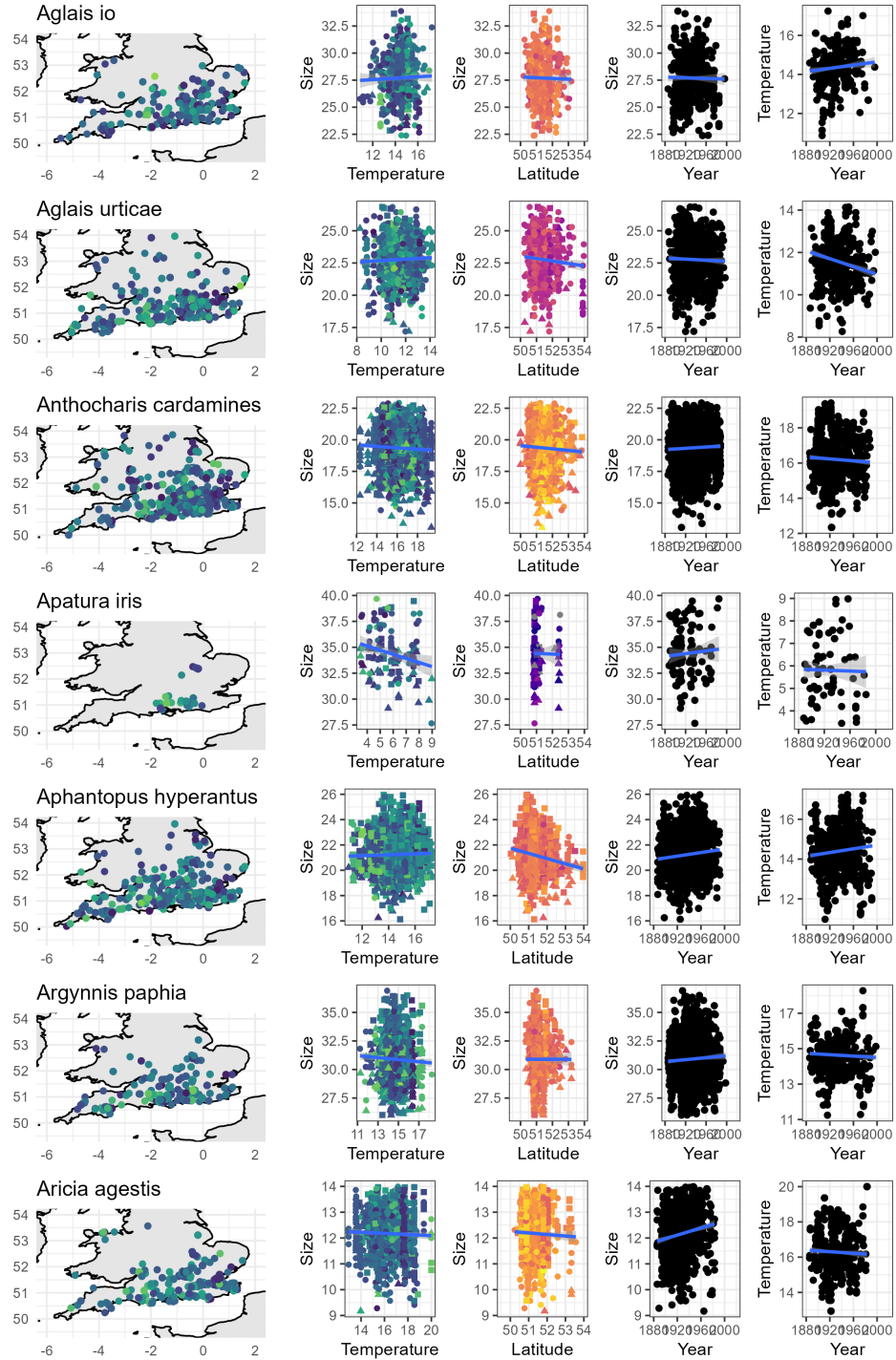
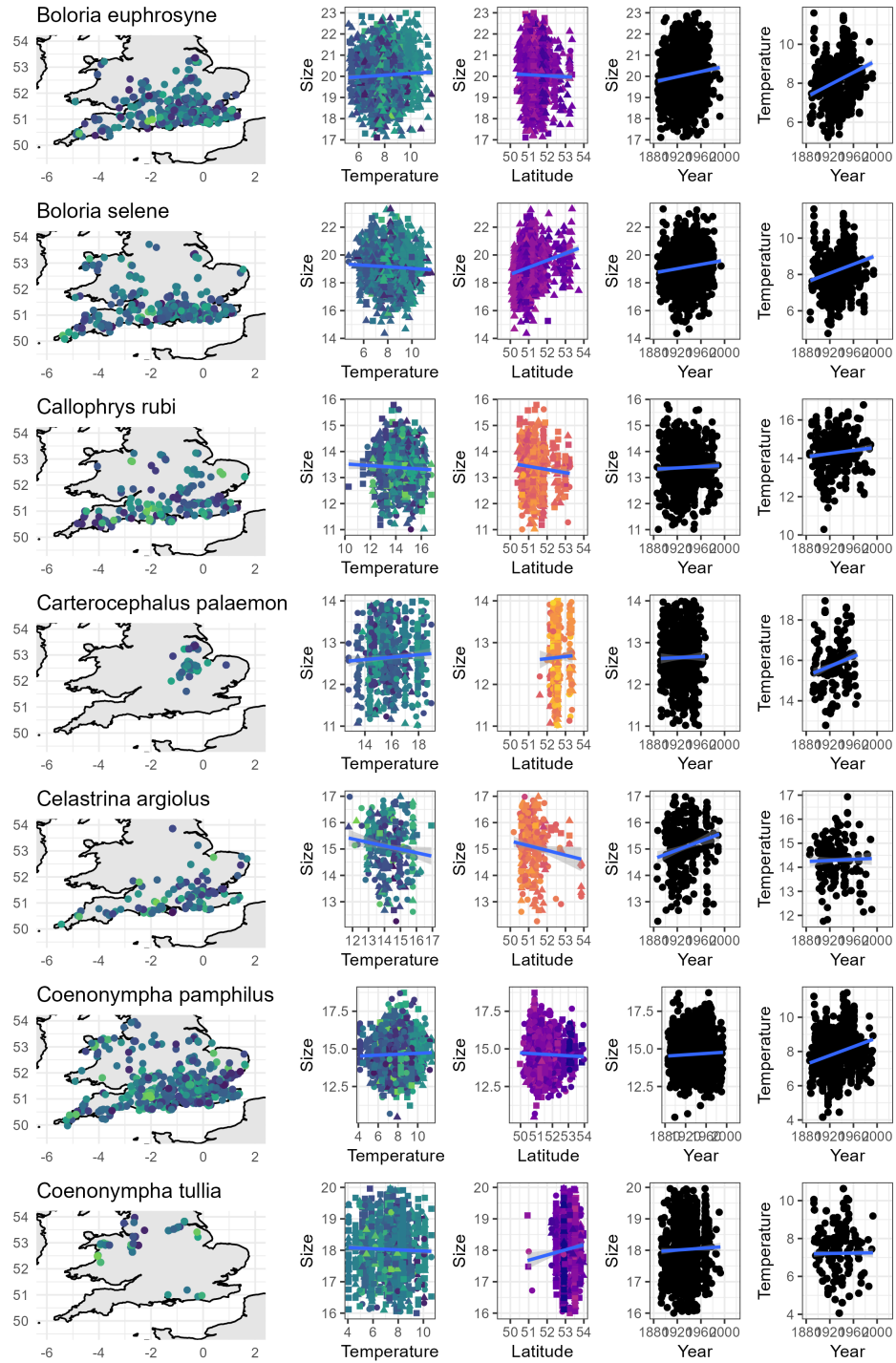
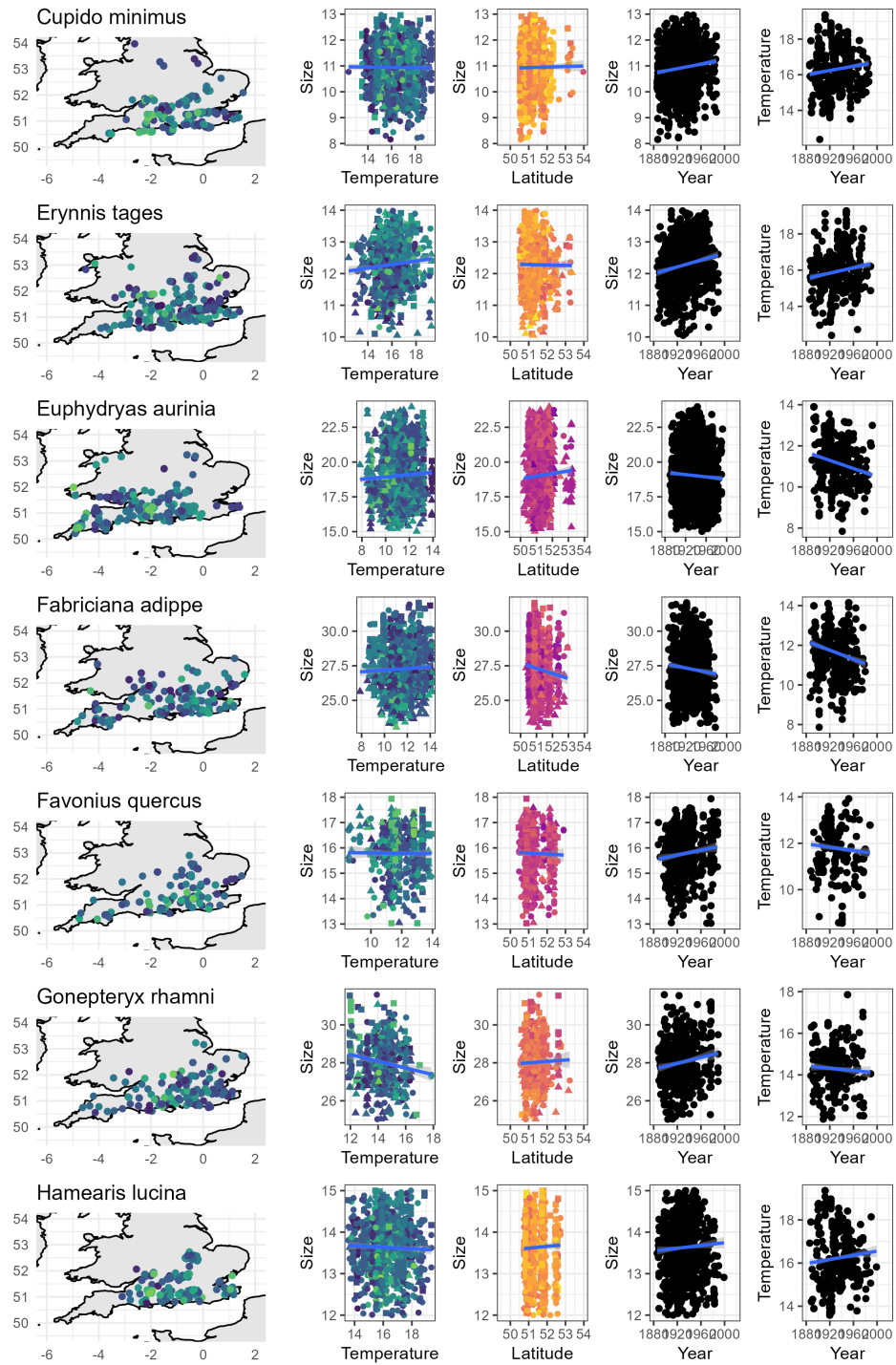


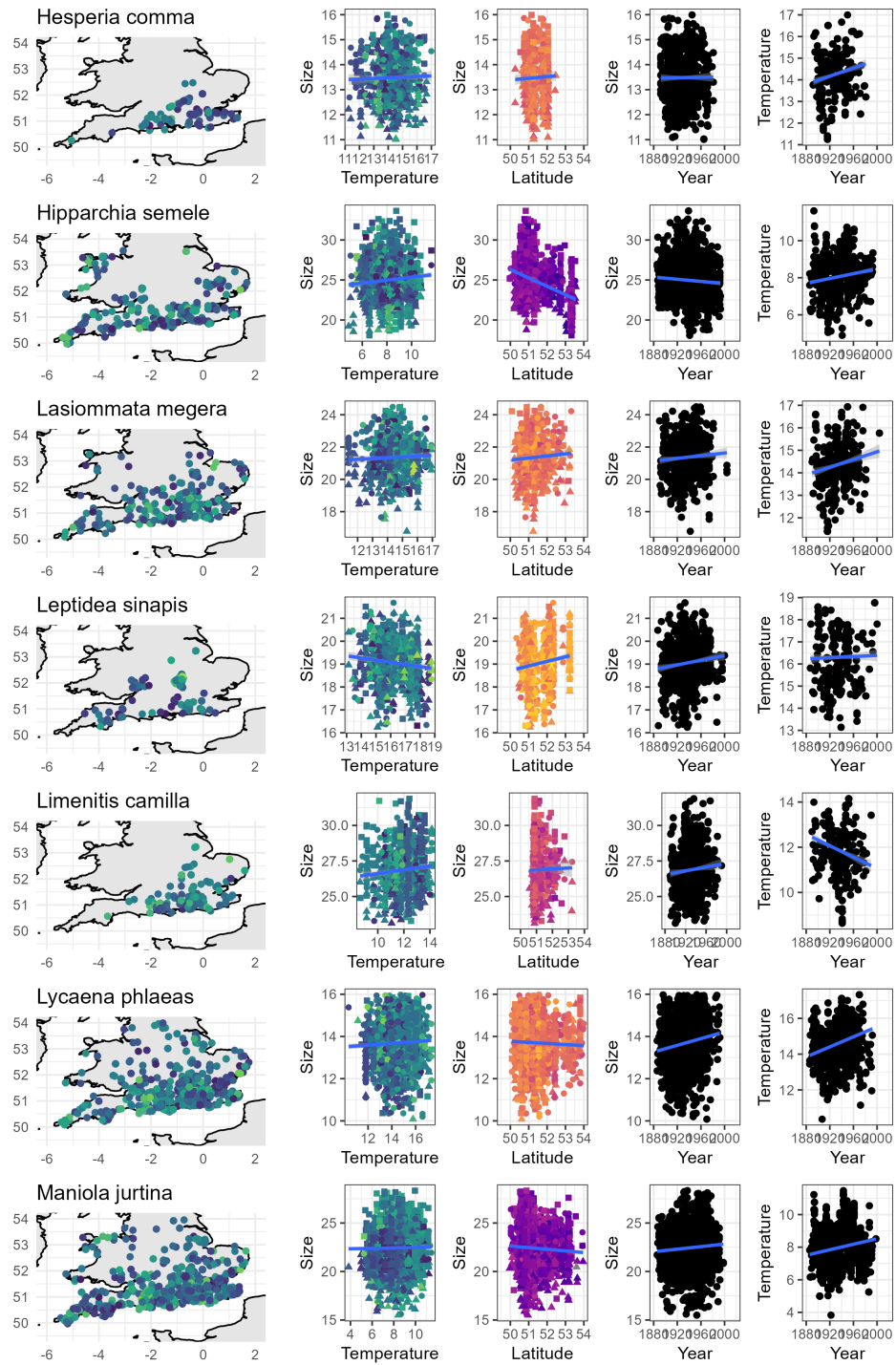
Figure S5: Coefficient Correlations between datasets. Here, each dot represents a species best-fit coefficient of the record-predictors within the two datasets. Error bars show 90% posterior confidence interval. Estimates from the smaller dataset were much more uncertain (note the unequal axis scales). Pearson's correlation of the mean estimates between the datasets was insignificant in each case (p-values: 0.158, 0.429, 0.0797 respectively (unadjusted for multiple tests)).

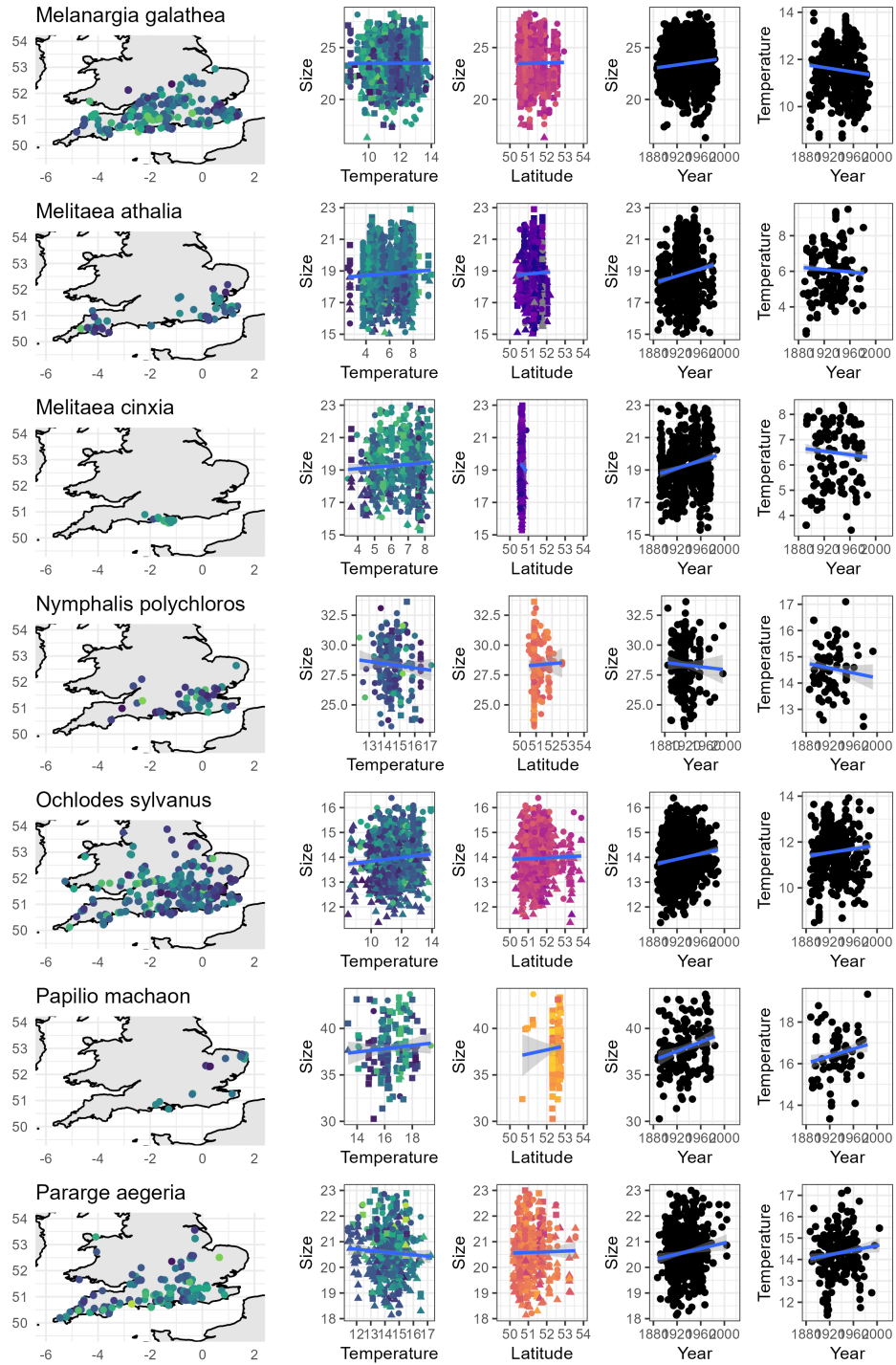
3 Specimen Spatio-temporal Distributions

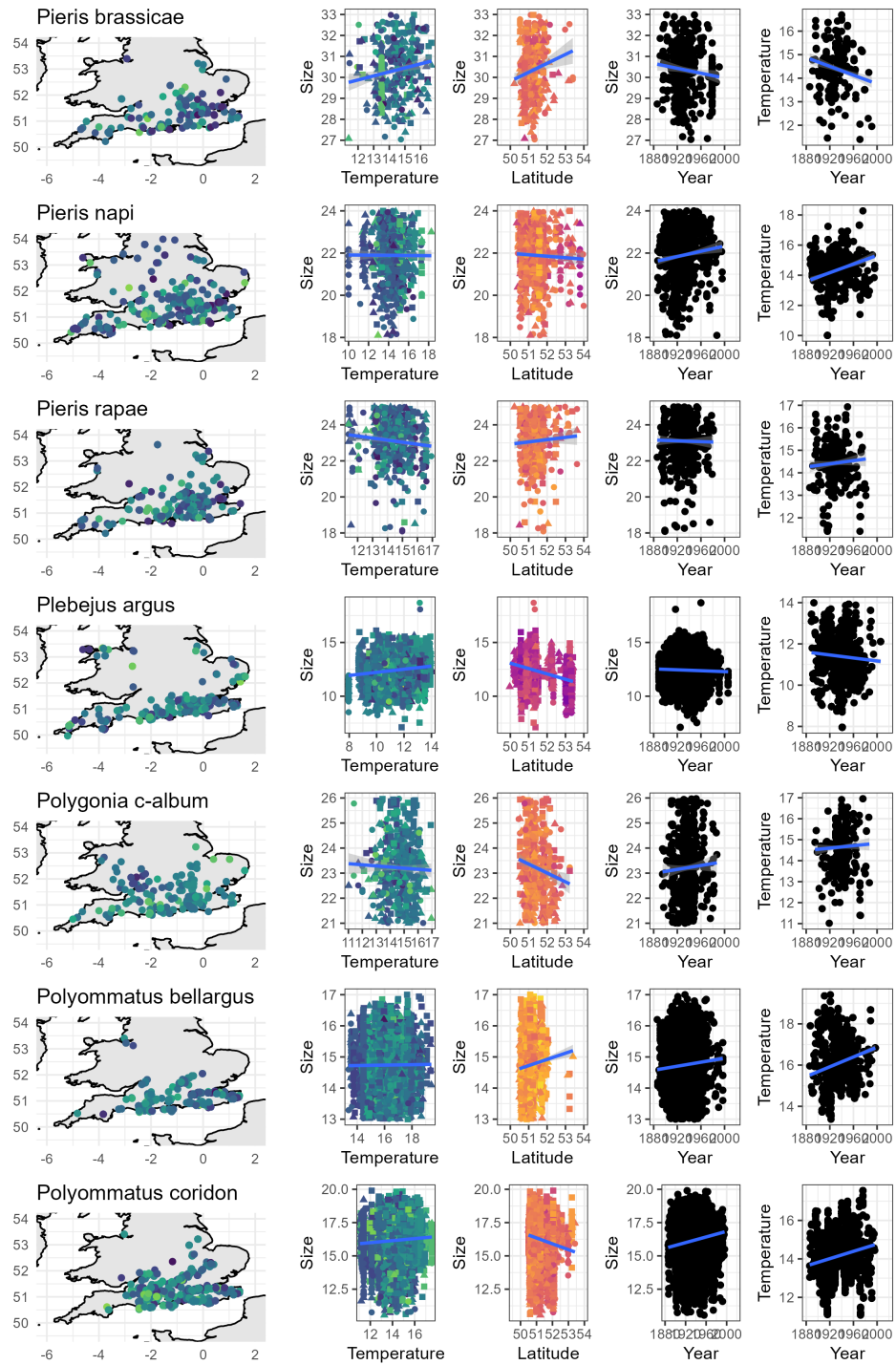


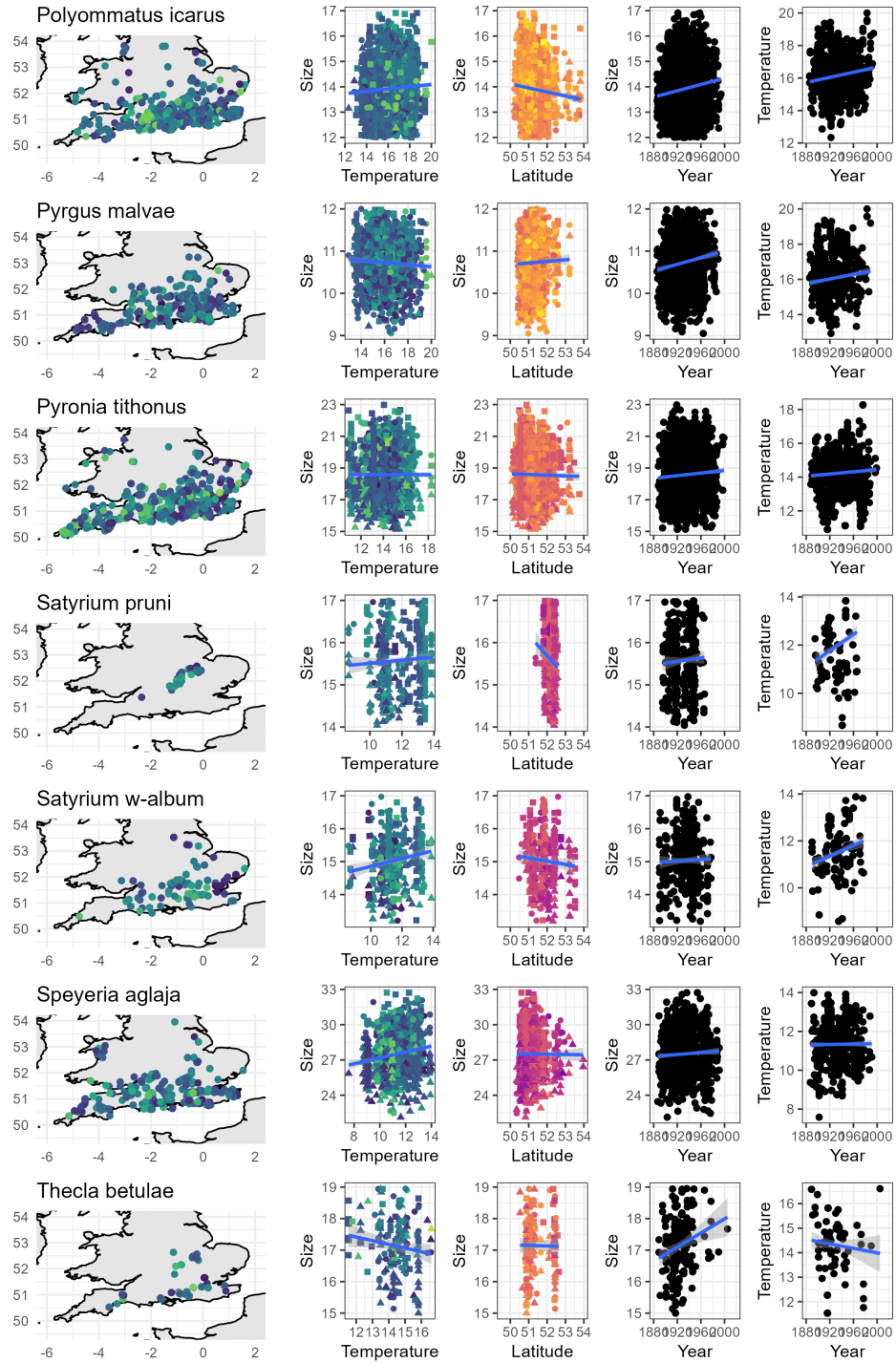












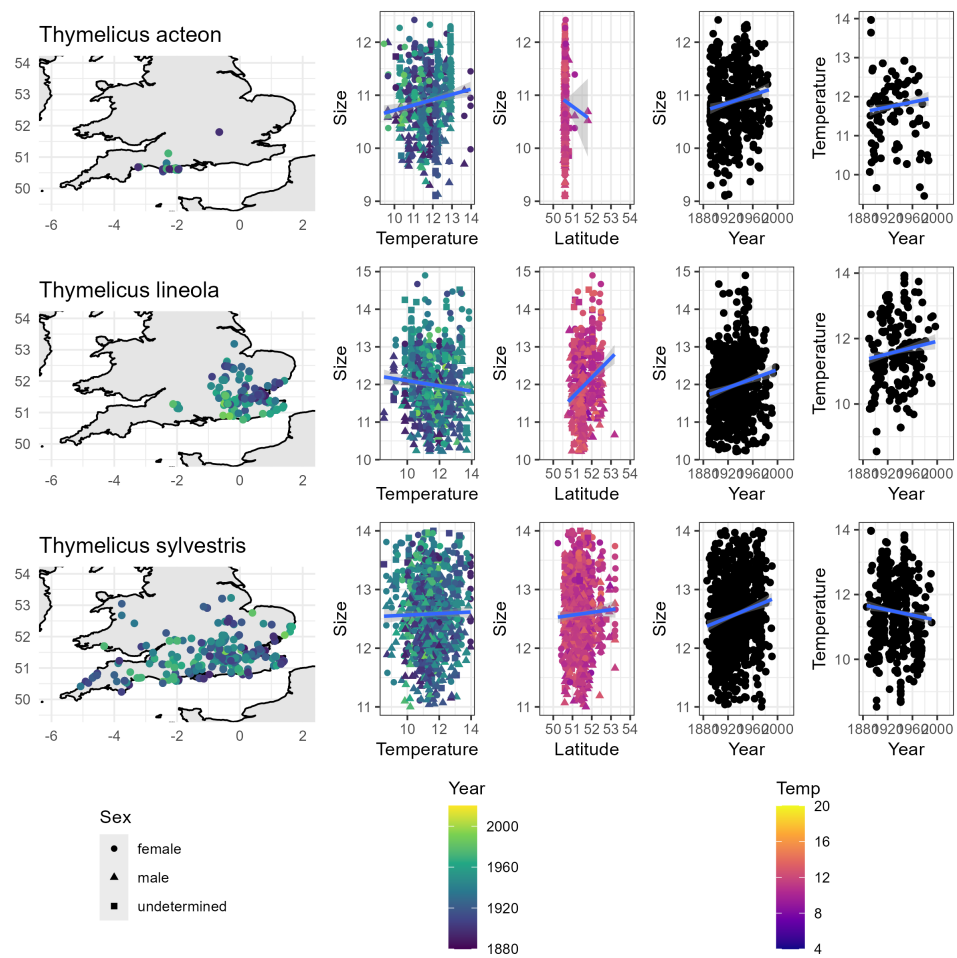
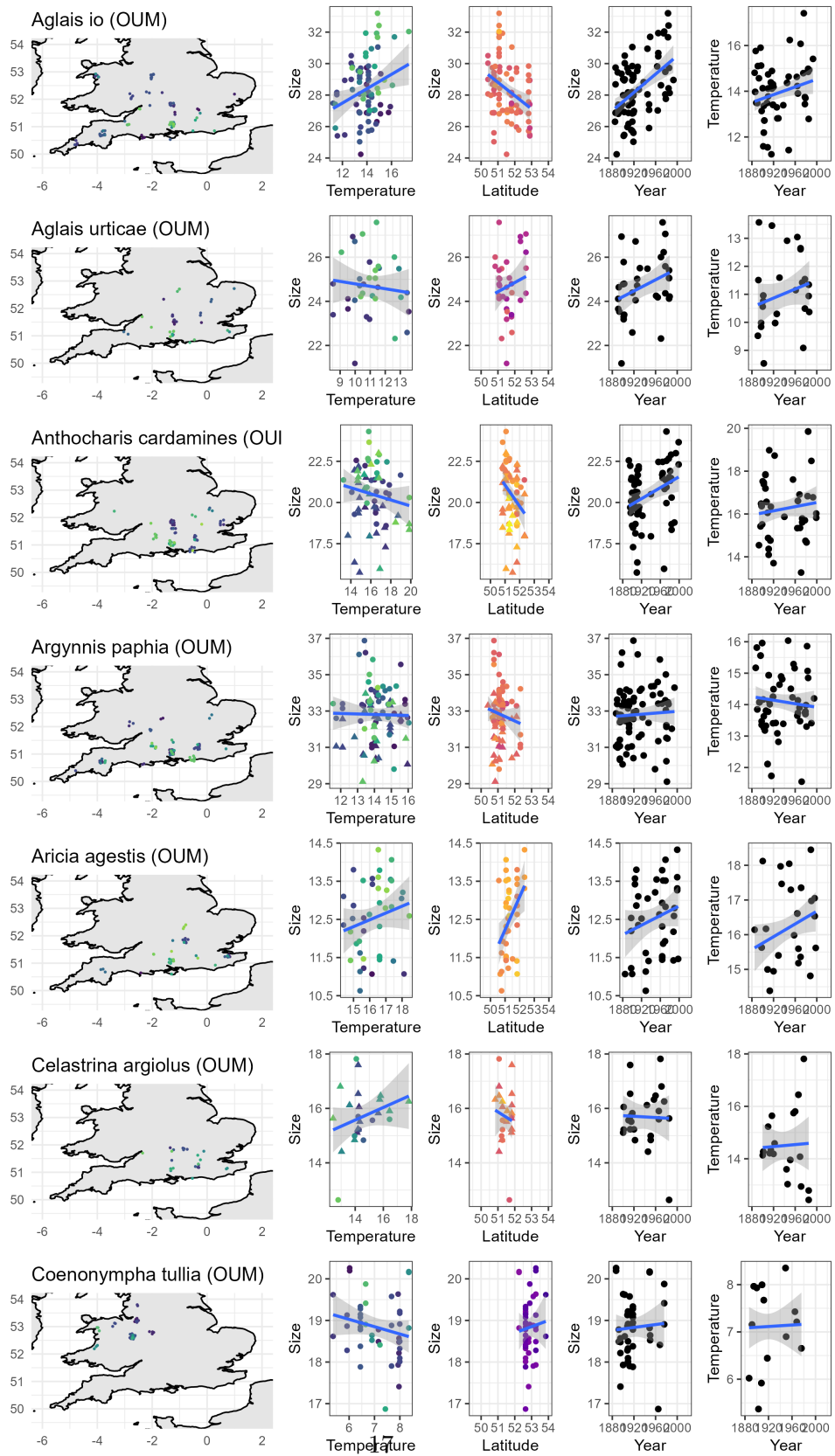
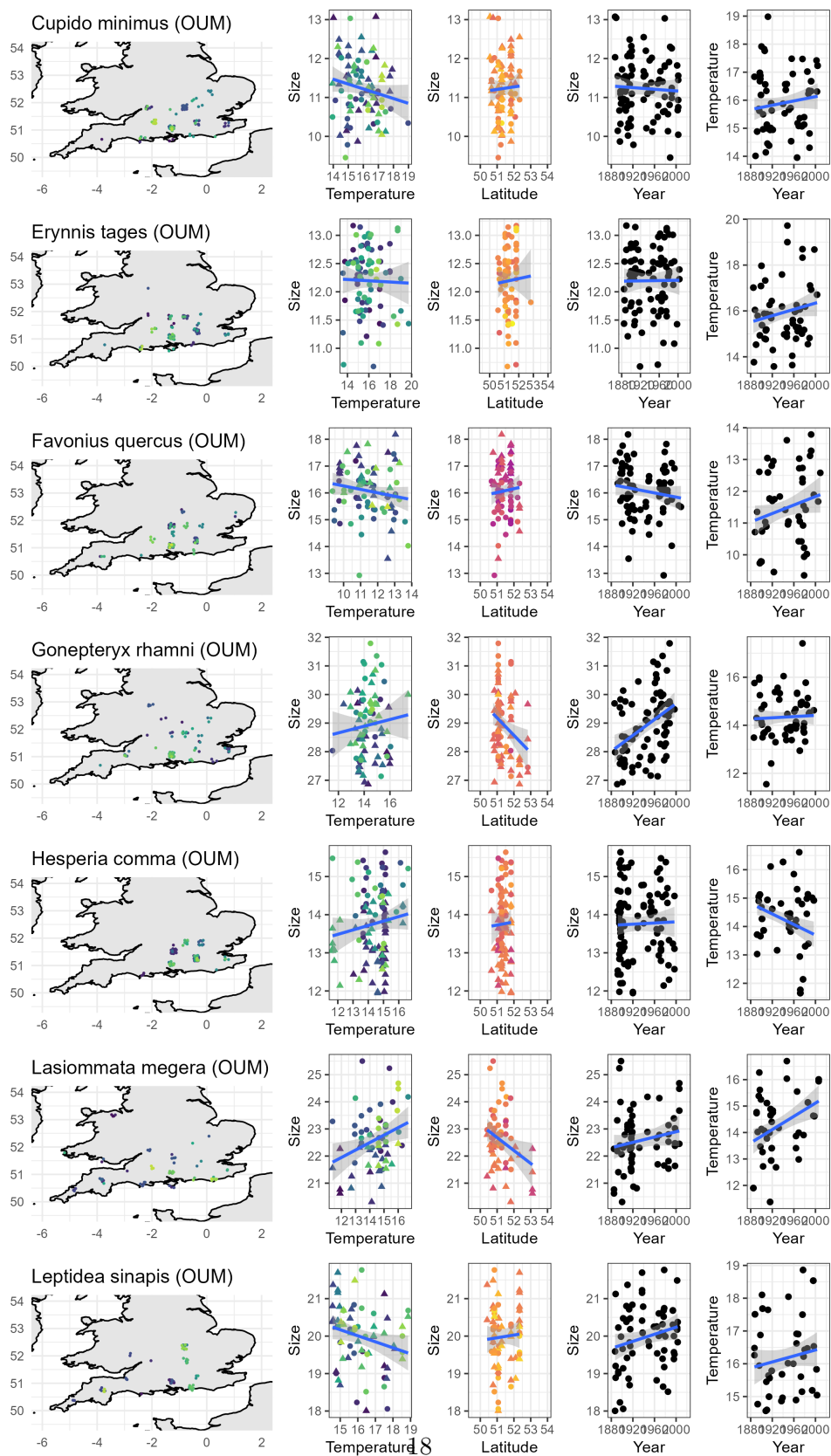
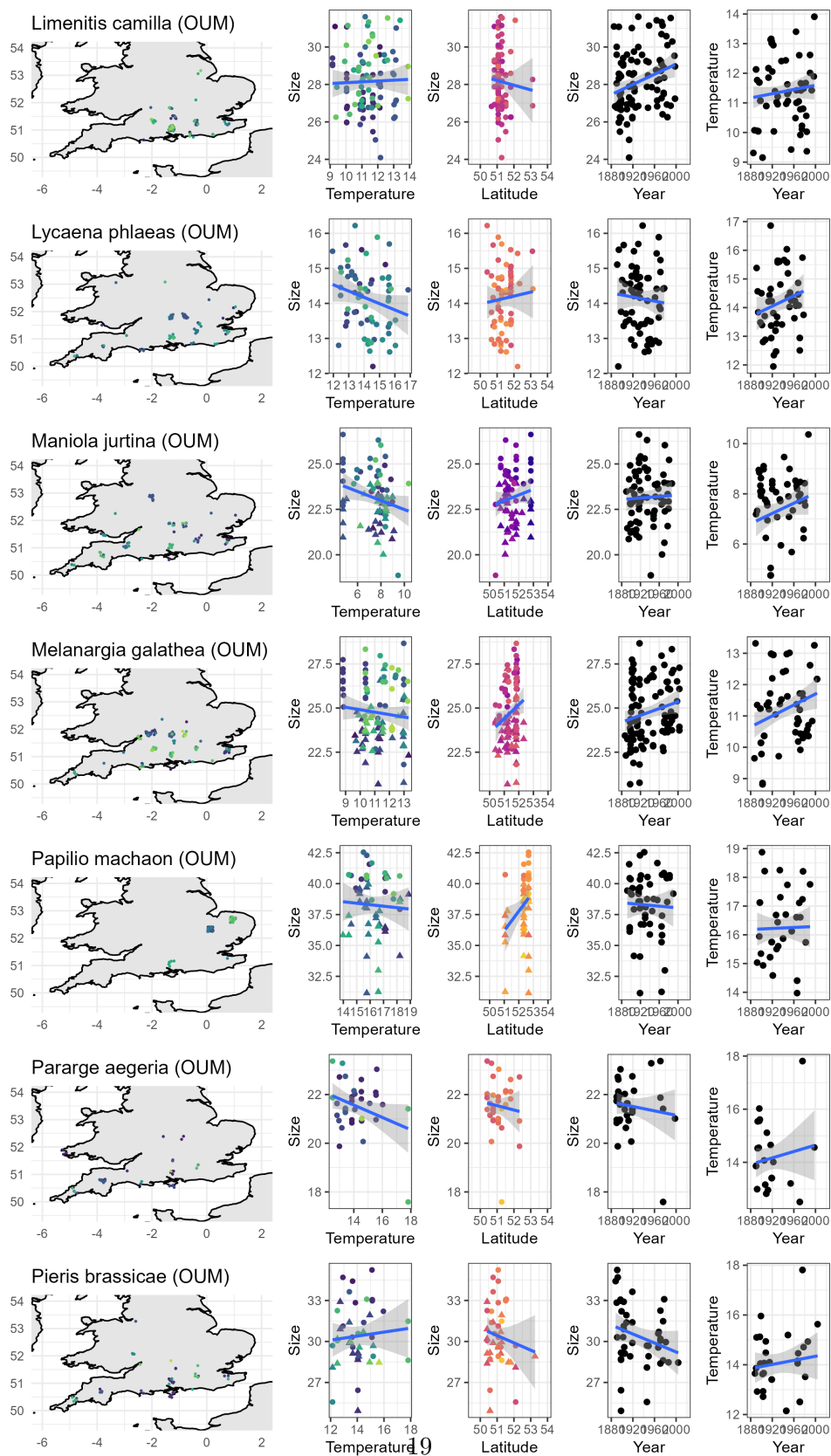


Figure S5: Distribution of records (post-filtering) from the NHM dataset for each species







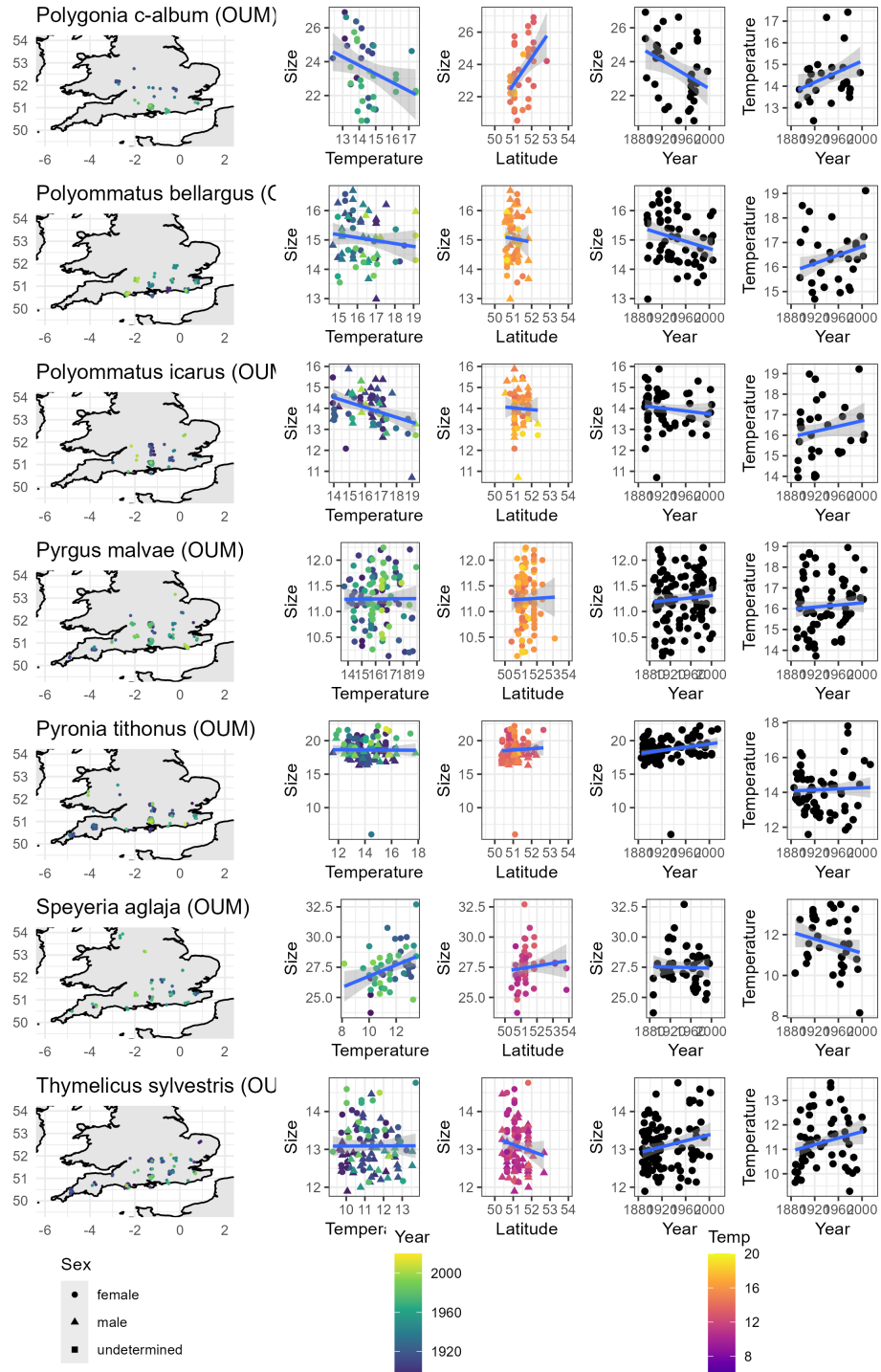


Figure S6: Distribution of records (post-filtering) from the NHM dataset for each species NB Locations on the map have been slightly jittered to reduce overlap.